AI Application R&D Engineer · BenQ Materials

Shan-Cheng Chang 張善程

LLM Agent Engineering · Knowledge Graph · Document AI

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— CORE SKILLS

Agent Engineering
Agent Harness · Function Calling · MCP · Skills / Tools · Context Management · Sandboxing · RBAC / Audit Log

Evaluation & Validation
Golden Set / Held-out · Code Validators · Regression Baseline · Failure Analysis

LLM & Retrieval
Gemini API · Azure OpenAI · Prompt Engineering · RAG · Knowledge Graph (Neo4j) · pgvector

Document AI / ML / CV
Azure Document Intelligence · OCR + LLM Extraction · XGBoost · SHAP · YOLOv8 · PyTorch

Backend & Web
Python · FastAPI · Node.js · React · TypeScript · PostgreSQL · SQL Server

DevOps & Tools
Docker · GCP · Azure · Git · Claude Code

Security
AES-256-GCM · JWT · AD / Entra ID · ISO 27001

— EDUCATION

National Taiwan University
M.S. Biomechatronics Eng. (AIoT / CV) · 2021 – 2023

National Taipei University of Technology
B.S. Mechanical Engineering · 2017 – 2021

— LANGUAGES

Mandarin Chinese — Native
English — Professional (TOEIC 790)
German — A2
Company Digital Talent L3 (Expert)
Military service completed

AI application engineer who designs, verifies and maintains enterprise AI systems in manufacturing: lead developer of BenQ Materials' on-prem AI data-analysis platform and Claude Code–style agent harness, in daily production use, after building the company's document-AI stack from zero. Every system ships with a labelled test set or regression check.

4.15M NTD	6	~1M NTD	1,500+ h
AUTOML PURCHASE AVOIDED / 2 YRS	PRODUCTION-LINE FLOWS ON PLATFORM	TANGIBLE BENEFIT / YR (3 OCR)	SAVED / YR (3 OCR SYSTEMS)

— EXPERIENCE

AI Application R&D Engineer — Dec 2024 – Present

BenQ Materials Corp. (明基材料) · Digital Technology Center, Intelligent Application Dept., Intelligent Application Development Sec. · Taoyuan, Taiwan

- **Enterprise Brain → AI data-analysis platform** (lead developer) — live on company servers and used daily by the BML polarizer line, replacing a purchased AutoML package (**4.15M NTD / 2 yrs**); 6 production-line report flows cut over with zero downtime. FastAPI + Gemini function calling + MCP, 10 Docker services.
  - **Architecture & trade-offs** — Claude Code subset with identical skill/permission formats; policy as deny rules, not prompts. 6 n8n workflows (~550 nodes) → 1 skill + 7 code flows; Excel uploads, the API and 6 scheduled flows all run the same sensor-analysis skill — statistics computed in code, the LLM writes interpretations from those stats (a failed LLM call never blocks the numbers). Chat waits on a knowledge loop: engineers annotate results → AI check against existing knowledge → approval → knowledge graph.
  - **Planning & context** — 10-round tool cap with forced summary; lazy-loaded skills (prompt holds only the index); 10-turn history window, LLM summarization past 12 turns or 24K tokens, truncation after 3 failed compressions.
  - **Tool failures & safety** — backoff retries, errors returned to the model, a verify step for failed tools and hallucinated paths; 4-role tool allowlists, LLM critic before sandbox runs, capability-dropped sandbox with CPU/memory/time limits, DB-trigger–enforced append-only audit log.
  - **Regression & ML** — migration verified by golden counts, live replays, byte-identical PDFs and snapshot regression; XGBoost yield model **F1 0.74 ± 0.05** (lot-grouped 5-fold; a random split's 0.92 was leakage).
- **Document-AI applications (OCR + LLM)** — built the company's platform from zero (Azure Document Intelligence + Azure OpenAI, Python/FastAPI, React, SQL Server) and its reusable OCR → LLM → validation → feedback pipeline; each system scored on a hand-labelled test set before go-live.
  - **Quotation OCR** — validity-date accuracy **51% → 100%** (29-document regression set; quote date 86%); 30 s → 10 s per document; 758+ documents processed.
  - **Taiwan & China invoice OCR** — **31/32** (5 key fields all correct, incl. handwritten) and **31/31** (10 fields); saves 1,447 h/yr; live at 3 sites.
  - **Health-check OCR platform** — **97.4%** field-level accuracy (42 reports × 46 fields), 216 reports processed; AES-256-GCM, AD + JWT, RBAC and audit logging to ISO 27001.
  - **APO-OCR prompt-optimization engine** — designed the eval-driven loop: 32-invoice golden set (24 train / 8 held-out, 3 leakage paths removed), exact-match code validator (LLM proposes, code scores), budget/loop/failure breakers; mentored an intern to ship it — **8/8** held-out, tying the hand-tuned prompt, for $1.44.
- **Google Cloud Platform — supplier questionnaire management system** (live) — React 19 + Cloud Functions + Firestore; 52 APIs, Entra ID login, built solo in 4.5 months, handed over with GCP training.
- **ESG carbon & energy platform refactor** — **44K → 6K lines**, 5 → 1 hosts, validated web upload instead of a personal-PC import.

Shift Supervisor (值班股長) — Jan 2024 – Nov 2024

China Steel Corporation (中國鋼鐵) · Cold-Rolling Mill · Kaohsiung, Taiwan

- **Ran shift operations** on the cold-rolled coil line, keeping production continuous at **95% yield**.
- **Data-driven anomaly analysis** — process parameters (rolling force, tension, speed, temperature) plus quality data; SPC, trend and correlation analysis to find root causes of coil defects and downtime.
- **Anomaly & defect reports** (Python / Excel) for daily production meetings — the domain base for the later production-line AI platform.

— RESEARCH & SELECTED PROJECTS

AIoT Pest Monitoring & Bee Pollination (M.S. thesis, NTU) — 2021 – 2023

Lead developer of greenhouse AIoT systems: YOLOv8 + StrongSORT bee tracking (*99.5% mAP*), Mask R-CNN pollen segmentation, deep-learning pest counting (*95.8%*), kriging-based climate control.
PyTorch · YOLOv8 · StrongSORT · Mask R-CNN · IoT · Kriging

Universal OCR Platform — no-code OCR builder (side project) — 2025 – 2026

Define fields and prompts, run a 7-step OCR → LLM pipeline, deploy as a REST API; three cloud variants (FastAPI · React 19 · Supabase · Cloud Run · Docker).

— PUBLICATIONS & PATENT

Chou, C.-Y., **Chang, S.-C.**, et al. (2023). Development of AIoT system for facility asparagus cultivation. *Computers and Electronics in Agriculture*, 206:107665. *(co-first author with advisor)*

Chang, S.-C., et al. (2022). AIoT based pest counting system for asparagus cultivation. *Proc. 10th ISMAB.*

Taiwan Invention Patent TW I842416 B — Crop Harvester, 2024.